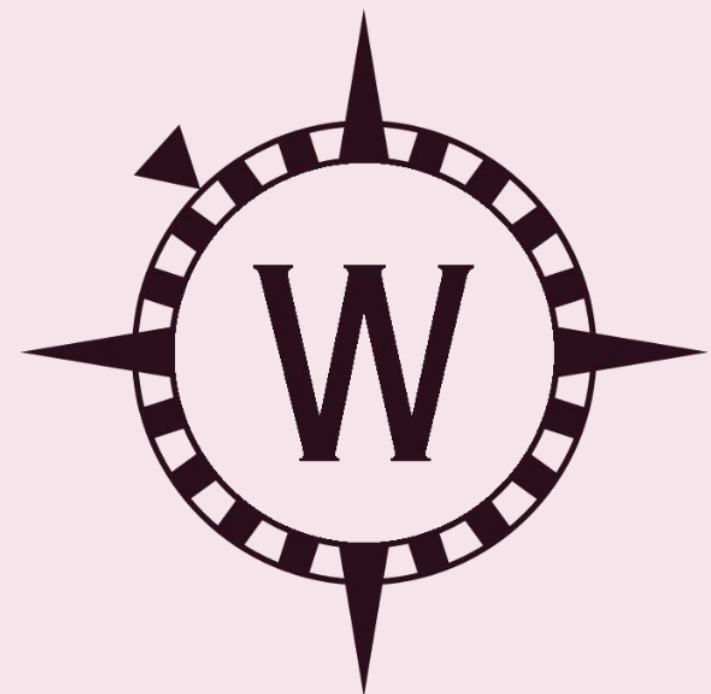




HEARD BUT NOT STUDIED

Mapping the gap between **health research** and women’s **lived experiences**.

Research Question

What women’s health **symptoms** and **conditions** are most discussed on **r/WomensHealth**, and to what extent are these topics represented in clinical research on **PubMed** and **ClinicalTrials.gov**?

Background & Related Work

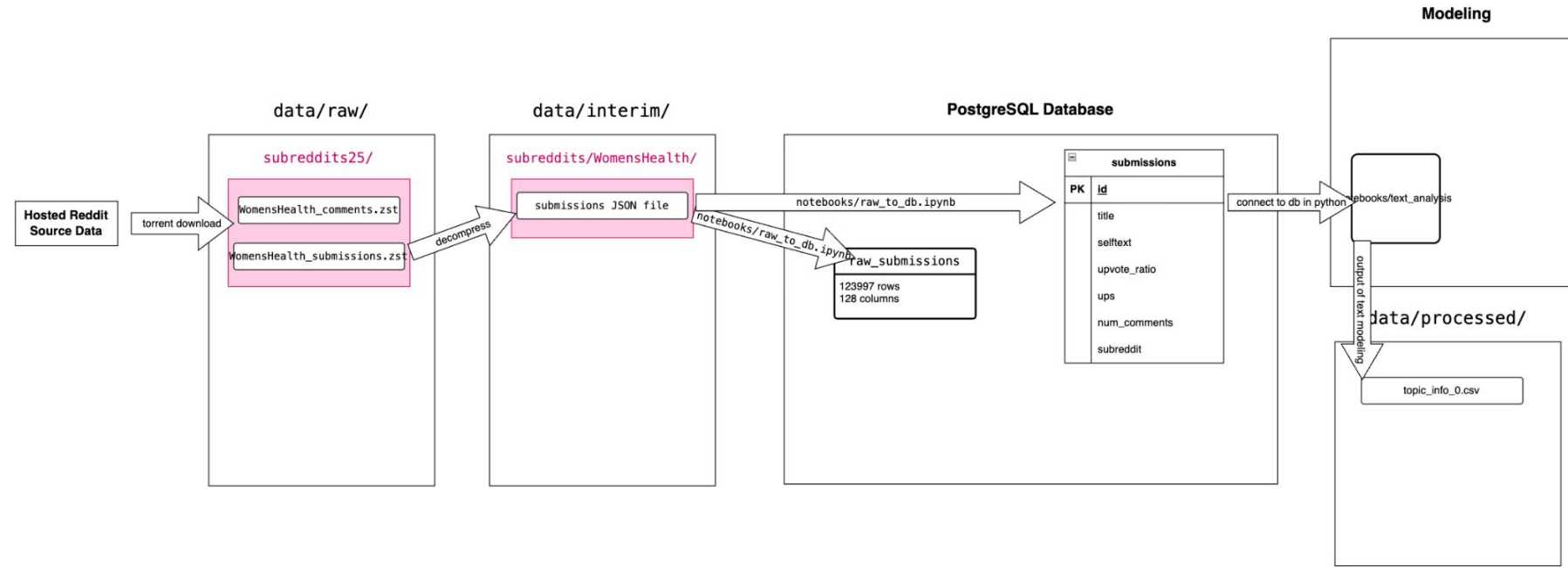
Previous studies used funding patterns and participant data to assess gaps in women’s health research. They found that:

- **Men’s health research gets 1.84x more funding**
A 2020 study looked at the funding of research towards male-dominant vs female dominant diseases by the U.S. National Institutes of Health (NIH). They found that in 2019, the NIH funded research for men’s diseases 1.84 times more than women’s.
- **7 out of 11 research areas underrepresented women**
A 2019 study quantified sex bias in clinical studies by analyzing participant data of common diseases that affect men and women. They found substantial underrepresentation of women in 7 of the 11 disease categories they analyzed.

However, there is very little research comparing women’s lived experiences with published health literature. Our project aims to help close this gap.

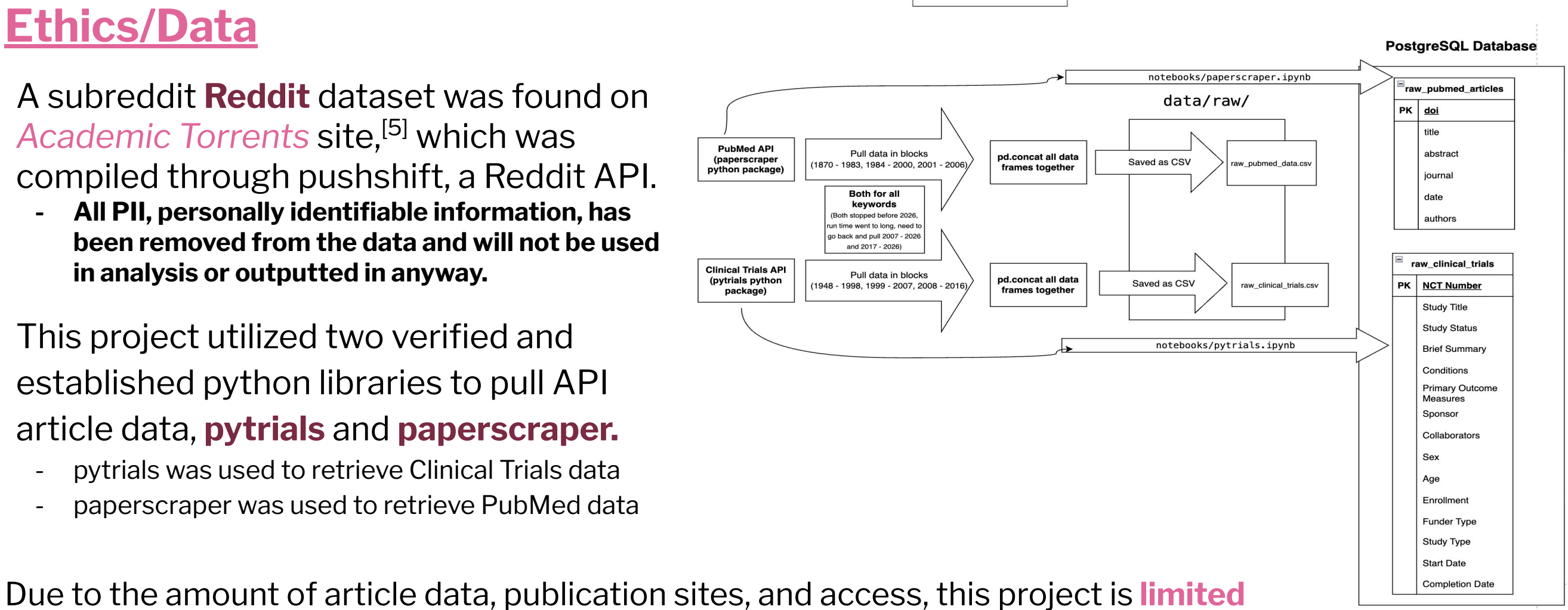
Data, Engineering, and Ethics

Reddit Data Pipeline



	Reddit	Clinical Trials	PubMed
Size	Rows: 123,997 Columns: 7	Rows: 373,282 Columns: 17	Rows: 10,206,788 Columns: 6
License	No license: Public domain access.	Free software under the BSD license	MIT License Copyright (c) 2020 PhosphorylatedRabbits

Article Data Pipeline



A subreddit **Reddit** dataset was found on **Academic Torrents** site,^[5] which was compiled through pushshift, a Reddit API.

- **All PII, personally identifiable information, has been removed from the data and will not be used in analysis or outputted in anyway.**

This project utilized two verified and established python libraries to pull API article data, **pytrials** and **paperscraper**.

- pytrials was used to retrieve Clinical Trials data
- paperscraper was used to retrieve PubMed data

Due to the amount of article data, publication sites, and access, this project is **limited** in an overarching accurate analysis of **everything researched** and **everything being discussed**.

This project also acknowledges the gap in research regarding the intersectionality of race and gender, and gender and sex, which is not covered in our analysis. Due to the limited data, this project is not able to accurately analyze individuals who face female sex specific issues outside the gender binary. Women of color face systemic components, including stereotyping, insurance access, and accessible healthcare, that all contribute to the imbalance of equity for racial groups within healthcare. Medical literature overall struggles to report data regarding sex and gender; it struggles even more so with reporting race and ethnicity.^[6]

Abstract*

★ ABSTRACT IS PLACEHOLDER

Women’s health research in _____ areas are _____ researched. It was found that of _____, _____, _____ topics most discussed by women on Reddit, PubMed and Clinical trials have _____ shown research among _____ topics and conditions.

Introduction & Motivation

Women’s Health Includes:

- Conditions of increased prevalence in women compared to men
- Pregnancy-related health disorders
- Gynecologic complications and cancers of specific tissues^[1].

Conditions pertaining exclusively to female reproductive organs like **endometriosis**, **uterine fibroids**, and **Polycystic Ovarian Syndrome** remain **underfunded** and **understudied** relative to their prevalence^[1]. Throughout history, males have been used as a baseline, and sex is not taken into account^[2]. Leading to a negative impact on women’s ability to understand their own health issues, receive diagnosis, treatment, and easily navigate healthcare systems.

A comparative analysis of experiential discourse from women and published medical understanding is crucial to map the gap in research.

Stakeholders

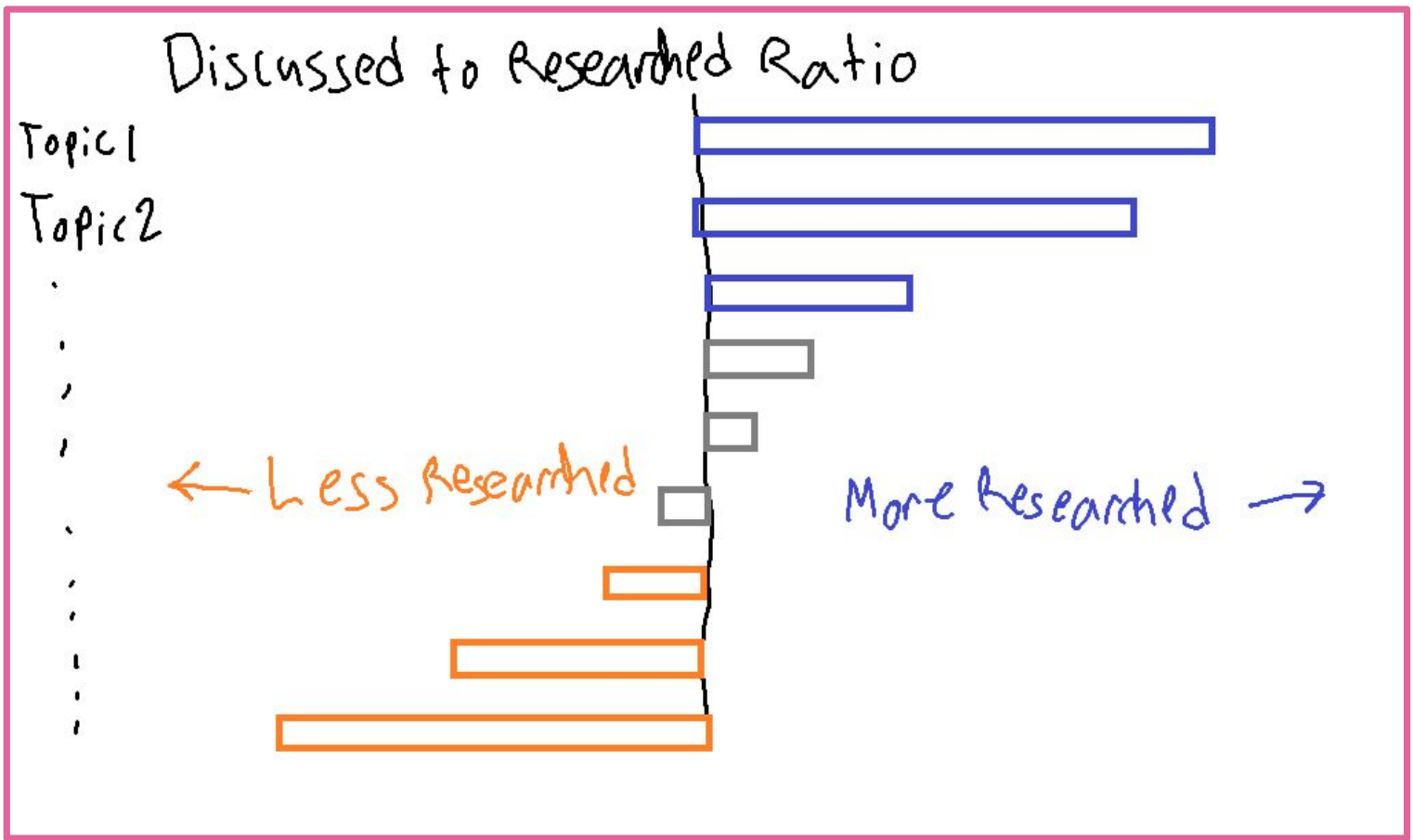
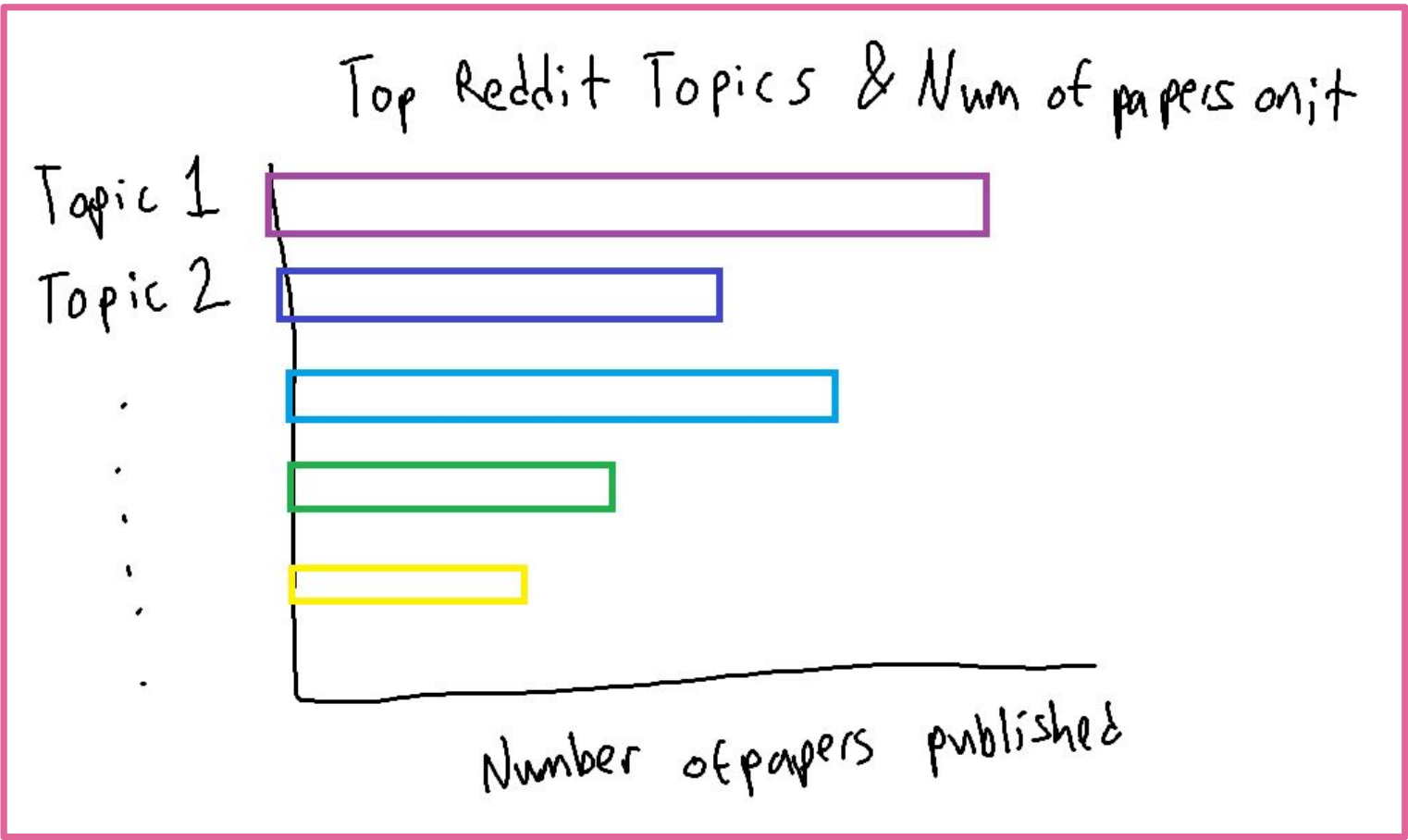
Primary stakeholder: Women

Who can make the change?

- Medical researchers
- Clinical trial designers
- Funders
- Public health policymakers

Secondary stakeholders include patient advocacy organizations who can use gap findings to strengthen funding arguments.

Results & Visualizations PLACEHOLDER



Reddit Data Topic Modeling

Top 10 words + count

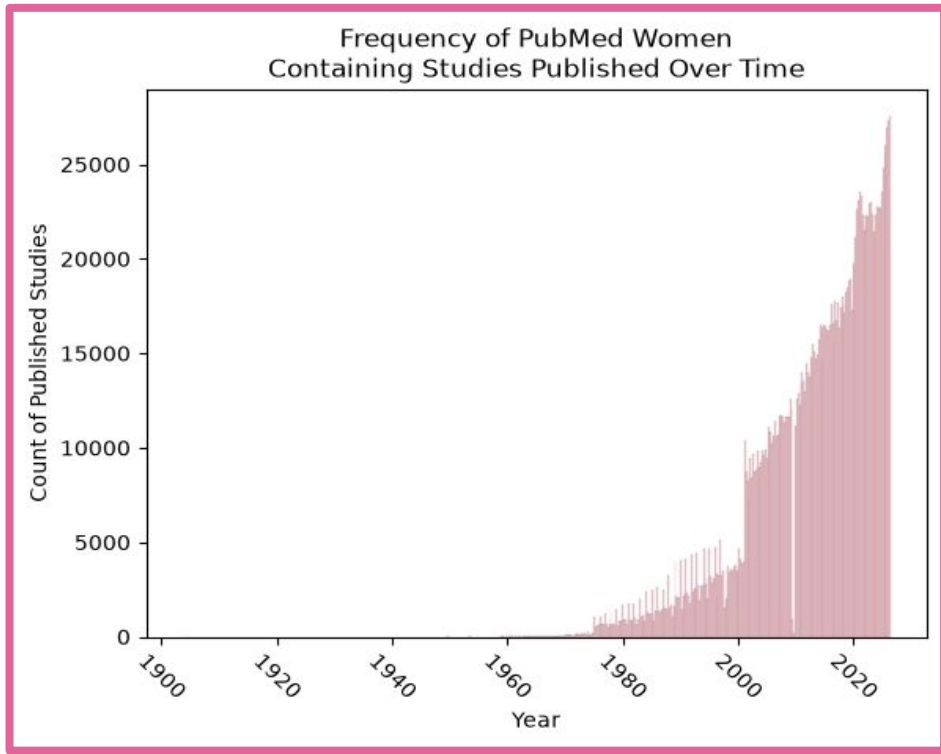
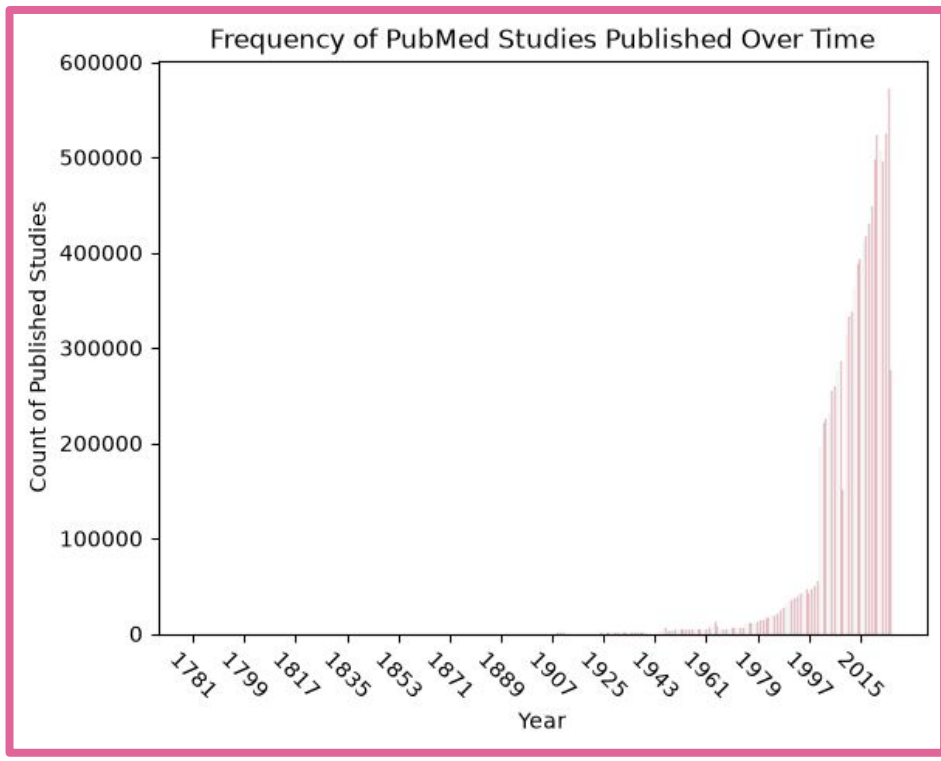
1. period: 50255
2. pain: 42155
3. doctor: 24154
4. normal: 24123
5. sex: 23973
6. symptoms: 17648
7. control: 17481
8. birth: 17472
9. bleeding: 15605
10. blood: 14916

Top 10 bigrams + count

1. birth control: 16173
2. yeast infection: 8502
3. pap smear: 2962
4. sexually active: 2960
5. side effects: 2398
6. yeast infections: 2318
7. ovarian cyst: 2062
8. boric acid: 1822
9. urgent care: 1593
10. pelvic pain: 1451

PubMed Studies Published Over Time

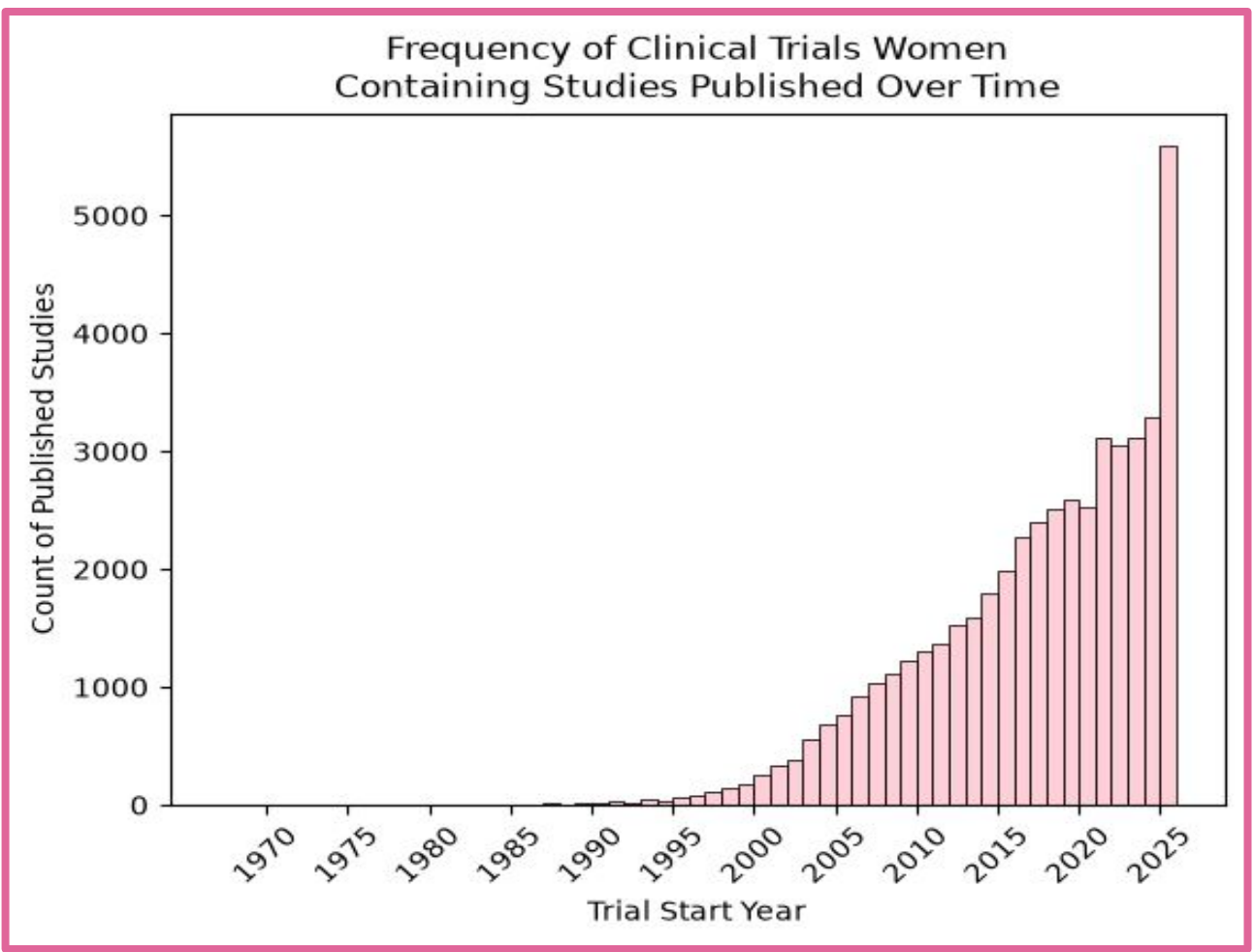
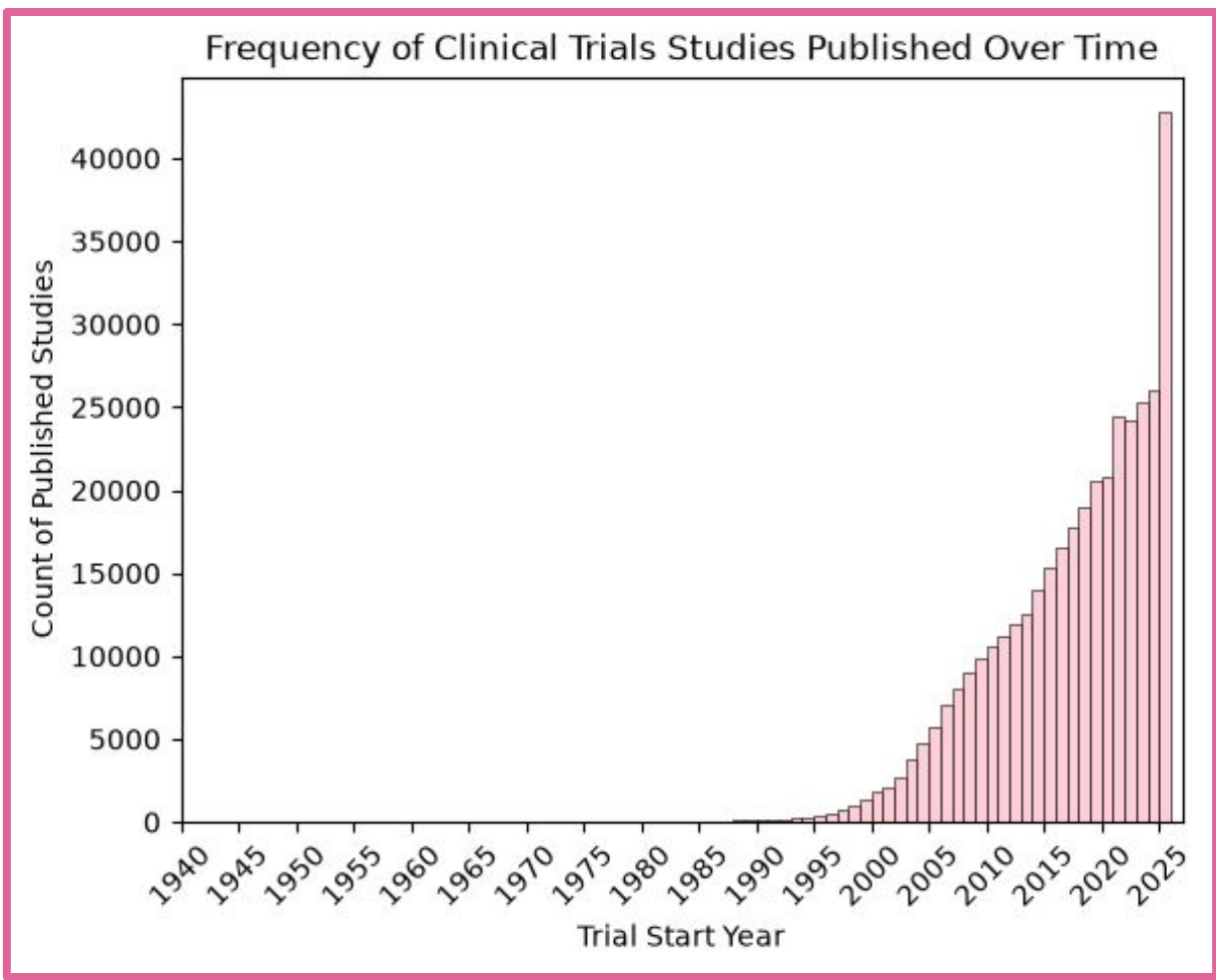
The chart to the right shows the development of research over time. The large number of papers in our dataset prove how vast research is, and the difficulty it poses to computational efforts.



The above figures show the gaps between the most discussed topics and how much that topic is researched. Topic 2 is more discussed than Topic 3, however, it has less research papers.

From the second chart, we can see that Topic x and y are severely under-researched. All numbers have a p-value of less than 0.001.

ClinicalTrials.gov Studies Published Over Time



These charts show the somewhat recent development of women’s health research, and the scales indicate women’s health have significantly less trials published.

Methods & Evaluations PLACEHOLDER

Frequency analysis: This is a computationally light way of gathering most commonly mentioned keywords, which is useful in assessing the topic makeup of r/WomensHealth at a face value, assist with data cleaning, and lead into the more advanced topic modeling steps.

Evaluation: Chi-square test or Log-likelihood ratio / G-test

BERTopic: A pre-loaded model that finds topic clusters, identifies keywords associated with a topic, and example text entries to assist the exploration and analysis of topics most discussed in the reddit data.

Evaluation: The following tests & expected good result:^[7]

- C_V: 0–1; higher is better (often 0.5–0.7+ after good preprocessing).
- UMass: ≤ 0; closer to 0 is better (very negative = poor co-occurrence).
- Silhouette: [–1, 1]; > 0.2 modest, > 0.5 strong (text often yields modest values).
- Diversity: [0, 1]; > 0.85 suggests distinct vocabularies, but check for over-fragmentation.
- Entropy (norm): [0, 1]; closer to 1 = balanced topic sizes; closer to 0 = dominance by few topics.

Baseline/Validation Strategy:

We will perform manual check on these calculations by performing random sampling and assessing those samples.

- “contains_women” column in article data
- Conditions column in Clinical Trials and keyword mapping column in PubMed
- Reddit - Article Data topic language

Frequency Results Baseline/Validation Strategy: PubMed, Clinical Trials, and Reddit data volume allows for higher generalizability across biomedical research and diverse demographic of online women communities. Due to the higher volume of data, topics that appear will have a higher statistical significance in being generalizable. To verify, a randomized sample of the Reddit data compared to the full sample will be compared, _____ topics stay the same.

Discussion*, Limitations, and Conclusion*

★DISCUSSION AND CONCLUSION ARE PLACEHOLDER

Discussion

These results mean there is a gap between what is posted about and what is researched. Online forums are biased, so while it does reflect what women experience, it does not mean every gap has been identified, and further research must be done.

Limitations

Our data sources and collection methods are limited, so the full scope of topics discussed and researched is not properly represented. Topic modeling and complex analysis is restricted by the computing power we have, so there is a limitation to the kinds of modeling and analysis we can perform, and with more in-depth data.

Conclusion

The results show significant gaps between research and what women experience for birth control, yeast infections, and ovarian cysts, with a p-value of _____, and _____, respectively. This means these are serious areas of women’s health that are under-researched. More funding and resources provided by organizations like the NIH should be allocated to these topics.

References & Reproducibility

Github Repository: <https://github.com/Addie-Shanti-Org/GenderDisparityinResearch>

Key dependencies: Python, PgAdmin, Railway, Pandas

References: [1] Edwards et al., Annu Rev Biomed Data Sci, 2023, [2] Karp, Mason, Beudet, et al., Nat Commun, 2017, [3] Mirin, A., Journal of Women’s Health, 2021, [4] Feldman, et al., JAMA Network Open, 2019, [5] Watchful1, Academic Torrents, 2026, [6] Bierer, et al., Cell Rep Med, 2022, [7] Pol, J., Network Is Life Github Pages, 2025